JanSamadhan Innovation Hub

An AI-powered platform connecting **citizens -> universities -> industry -> government** to turn real societal problems into practical, measurable solutions.

SIH Details	Information
Organization	Government of Jharkhand
Department	Department of Higher & Technical Education
Category / Theme	Software / Smart Education
Capstone Positioning	Explainable AI-driven problem intelligence, stakeholder matching and innovation lifecycle management platform

1. Problem We Are Solving

People in communities identify problems every day - poor water management, agricultural issues, sanitation, accessibility, education, healthcare, environment and local infrastructure. However, there is no common platform that converts these problems into structured innovation projects and connects them with the right people who can solve them.

At the same time, universities have students, faculty, laboratories and research expertise, while companies and startups have technology, funding, mentorship and implementation capabilities. These resources are often disconnected from the actual problems faced by communities.

The core gap: Problem discovery happens in one place, expertise exists somewhere else, and implementation support exists somewhere else - but there is no single intelligent pipeline connecting them.

2. Enhanced Core Idea

We propose a platform where a citizen or organization submits a real societal problem with description, photos/videos, location and supporting information. Government validates the report. AI then structures the problem, classifies it, identifies duplicates, estimates priority, extracts required expertise and recommends suitable universities using an explainable matching score.

Universities accept validated challenges, form student/faculty teams and develop solutions. Industry and startups can contribute mentorship, technology, funding, testing or deployment support. The platform tracks the complete innovation journey and measures real-world impact.

REPORT -> VALIDATE -> AI UNDERSTAND -> CLUSTER -> PRIORITIZE -> MATCH -> BUILD -> PILOT -> DEPLOY -> MEASURE IMPACT

3. How the Platform Will Work

Step	What Happens
1. Citizen reports	User submits a problem with description, category, location and multimedia evidence.
2. Government validates	Authorized officer checks whether the problem is genuine, useful and actionable.
3. AI understands	AI extracts domain, sub-domain, severity, urgency, affected population, required expertise and key entities.
4. Duplicate clustering	Embeddings + pgvector identify similar reports and cluster repeated community problems.
5. Priority intelligence	Explainable scoring recommends priority; government can accept, modify or reject it.
6. University matching	Hybrid semantic + weighted capability scoring ranks suitable universities/faculty with reasons.
7. Team formation	University creates a multidisciplinary student + faculty team and submits a proposal.
8. Industry matching	Relevant companies/startups/CSR organizations are recommended for mentorship, funding or technology.
9. Project execution	Team develops prototype, tracks milestones and advances through Innovation Readiness Levels.
10. Pilot & deployment	Solution is tested with the community and can move toward actual deployment.
11. Impact tracking	Platform records beneficiaries, before/after indicators, feedback and measurable improvement.

4. Main Users

User	Purpose
Citizens / Communities	Report problems, track status and provide feedback.
Government	Validate, prioritize, assign, monitor, audit and evaluate challenges.
Universities	Maintain capability profiles, accept challenges and build student/faculty teams.
Students	Work on real-world projects, prototypes, evidence and milestones.
Faculty	Mentor teams and technically review proposals and results.
Industry / Startups	Provide expertise, funding, technology, equipment, testing and deployment support.
Platform Admin	Manage roles, governance, taxonomies, security and system configuration.

5. Simple Example

Suppose several villages report recurring drinking-water failures. Instead of treating every report as an isolated complaint, the platform can recognize a larger regional problem.

- **AI:** identifies Water Management / Rural Drinking Water and extracts severity, urgency and estimated population affected.
- **Duplicate engine:** finds highly similar reports and groups them into one problem cluster.
- **Government:** validates the cluster and reviews the AI priority recommendation.
- **Matching engine:** recommends universities with water-resource, civil engineering, IoT and GIS capabilities and explains every match.
- **University + industry:** form a team, build a monitoring solution and run a community pilot.
- **Impact dashboard:** compares incidents before and after deployment and shows beneficiaries, improvement and citizen satisfaction.

6. Key Features - Enhanced Capstone Scope

A. Citizen Problem Portal

- Easy problem submission with text, photos, videos, documents and geolocation.
- Problem status tracking from report to deployment and impact.

B. AI Problem Intelligence Engine

- Automatic domain/sub-domain classification.
- Structured extraction of severity, urgency, affected population, required expertise, entities and keywords.
- Confidence score and evidence for important AI outputs.

C. Semantic Duplicate Detection & Problem Clustering

- Embedding-based similarity search using PostgreSQL + pgvector.
- Groups repeated reports into larger problem clusters and estimates total affected population.

D. Explainable Priority Intelligence

- Weighted priority score based on severity, population affected, urgency, public safety and recurrence.
- Government can accept, modify or reject the recommendation; changes are recorded.

E. Explainable University Matching

- Ranks universities instead of broadcasting every challenge.
- Uses faculty expertise, departments, labs, previous projects, success history and geographical feasibility.
- Shows a transparent score breakdown and recommendation evidence.

F. University Project Workspace

- Accept challenge -> create team -> assign faculty -> submit proposal -> track milestones.
- Maintain prototype links, documents, results and Innovation Readiness Level.

G. Intelligent Industry Collaboration

- Recommends relevant companies/startups/CSR organizations based on project needs.
- Supports mentorship, funding, APIs, equipment, testing and deployment assistance.

H. Government Intelligence Dashboard

- District/domain heatmaps, open challenges, high-priority clusters, project status and deployments.
- Audit trail for important decisions and AI-to-human overrides.

I. Social Impact Measurement

- Tracks beneficiaries, problem reduction, citizen satisfaction, sustainability and deployment success.
- Generates an explainable Impact Score rather than marking a project only as completed.

7. AI Problem Intelligence - Structured Understanding

The AI layer should visibly convert unstructured citizen reports into structured, reviewable information. AI remains an assistant; authorized humans retain final decision authority.

Example input: "For the last three months, our village hand pumps stop working every few days and nearly 600 people depend on them."

AI Output	Example
Domain / Sub-domain	Water Management / Rural Drinking Water
Problem Type	Infrastructure Failure
Severity / Urgency	High / High
Estimated Affected Population	~600
Required Expertise	Civil Engineering, Water Resources, IoT Monitoring, GIS
Keywords	hand pump, groundwater, rural infrastructure, water monitoring
AI Confidence	92%

8. Explainable Priority Intelligence

Priority should be calculated through a transparent scoring model instead of a hidden AI label.

Factor	Weight	Example Score
Severity	30%	26 / 30
Affected Population	25%	19 / 25
Urgency	20%	17 / 20
Public Safety	15%	12 / 15
Problem Recurrence	10%	8 / 10
Total	100%	82 / 100 -> HIGH

Human-in-the-loop: AI recommends HIGH -> Government officer can Accept, Modify or Reject. The final decision and reason are stored in the audit trail.

9. Semantic Duplicate Detection & Problem Clustering

Multiple citizen reports can represent the same root problem. Semantic embeddings enable the system to detect meaning-level similarity rather than relying only on matching words.

Related Report	Similarity
Village B water failure	93%
Village C pump outage	89%
Village D drinking-water interruption	84%

Cluster result: Rural Water Infrastructure Failure - combined estimated affected population: 4,350 citizens.

Technical flow

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Citizen Report -> Text Embedding -> pgvector Similarity Search -> Threshold / Top-K Retrieval ->
Duplicate Decision -> Problem Cluster
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10. Explainable University Recommendation Engine

This should be the main technical centerpiece of the capstone. The system should not only recommend a university; it should explain why that institution is suitable.

Matching Factor	Weight	Example
Faculty Expertise Similarity	30%	27 / 30
Relevant Department	20%	18 / 20
Previous Project Similarity	20%	17 / 20
Laboratory Capability	15%	13 / 15
Previous Success / Deployment	10%	9 / 10
Geographical Suitability	5%	4 / 5
Total Match	100%	88 / 100

Why recommended? Civil Engineering Department + Water Resources Research Lab + IoT Systems Lab + relevant faculty + previous water-management projects + prior rural deployment experience.

11. Evidence-Backed AI Explanations

Every important AI recommendation should expose the evidence used to produce it. This prevents the system from appearing as a black box and makes recommendations easier to validate.

Problem Evidence	Institution Evidence
"requires water monitoring and IoT expertise"	"Water Resources Research Centre"
"rural infrastructure failure"	"Department of Civil Engineering"
"continuous monitoring needed"	"IoT Systems Laboratory"

12. Innovation Readiness Level (IRL)

Every project progresses through a common innovation maturity model so all stakeholders can see how close a solution is to real deployment.

Level	Meaning
IRL-0	Problem reported
IRL-1	Problem validated
IRL-2	Research completed
IRL-3	Concept developed
IRL-4	Prototype developed
IRL-5	Prototype tested
IRL-6	Community pilot
IRL-7	Deployment ready
IRL-8	Deployed
IRL-9	Impact verified

13. Social Impact Measurement Engine

The project should measure whether a solution actually improved the real problem. Success must go beyond "project completed".

Metric	Example
People affected before solution	4,350
People benefiting after deployment	3,920
Problem frequency before	17 incidents / month
Problem frequency after	3 incidents / month
Measured improvement	82.4%
Citizen satisfaction	4.3 / 5
Deployment status	Active
Impact Score	88 / 100

Example Impact Score

Factor	Weight
Population Benefited	25%
Problem Reduction	30%
Citizen Satisfaction	20%
Solution Sustainability	15%
Deployment Success	10%

14. Geospatial Intelligence

Location data should power government decision-making, not merely display a pin on a map.

- State/district heatmaps by problem domain and severity.
- Cluster visualization for repeated community issues.
- District drill-down: open problems, high-priority problems, projects running, pilots, deployments and resolved cases.
- Location-aware university and industry matching where geography is relevant.

15. Security, RBAC & Auditability

The platform contains multiple roles and important public-sector decisions, so security and traceability should be treated as first-class capstone requirements.

Security Control	Implementation Direction
Authentication	JWT or secure server-side session
Authorization	RBAC + resource-level permission checks
Passwords	Strong hashing + secure credential handling
Uploads	File type, size and malware-safe handling controls
APIs	Input validation, rate limiting and authorization
Sessions	Expiration and secure cookie/token handling
Audit Logs	Record validations, overrides, assignments and status changes
Data Minimization	Collect only information needed for the workflow

16. Proposed Technical Architecture

The capstone should remain a modular web application rather than an unnecessarily complex microservice system. Technical depth should come from the intelligence, explainability, evaluation and workflow quality.

Component	Recommended Technology
Frontend	React.js / Next.js
Backend	Go (net/http) or Node.js / Express
Database	PostgreSQL
Similarity Search	PostgreSQL + pgvector
File Storage	S3-compatible object storage
AI / ML	Embedding model + classification / extraction service
Matching	Semantic similarity + weighted structured scoring
Authentication	JWT / secure session + RBAC
Maps	OpenStreetMap-based solution / suitable map API
Deployment	Docker + cloud / VPS
Observability	Application logs + audit events + basic performance monitoring

17. Target MVP - What We Will Demonstrate

Citizen submits problem -> Government validates -> AI structures & categorizes -> Duplicates clustered -> Priority explained -> University matched -> Team created -> Proposal submitted -> Milestones tracked -> Pilot/deployment recorded -> Impact measured -> Dashboard updated.

- Role-based login for Citizen, Government, University/Faculty/Student, Industry and Admin.
- Working challenge submission with multimedia evidence and geolocation.
- AI structured extraction + classification with confidence.
- Semantic duplicate detection and problem clustering.
- Explainable priority scoring.
- University capability profiles and explainable matching.
- Project/team/milestone management with IRL.
- Industry collaboration request / matching.
- Government geospatial analytics dashboard.
- Impact measurement and audit trail.

18. What We Will NOT Overbuild in the MVP

To keep implementation strong, the following should remain future extensions unless the core pipeline is fully complete: nationwide integrations, advanced multilingual offline mode, patent/technology-transfer management, complex external grievance-system integrations and large-scale predictive models.

19. Evaluation Methodology - Academic Strength

The AI modules must be measured. This is the difference between "we used AI" and a technically defensible capstone.

AI / System Module	Evaluation Metrics
Problem classification	Accuracy, Precision, Recall, F1
Structured extraction	Field-level accuracy / exact-match rate
Duplicate detection	Precision, Recall, F1
Similarity retrieval	Precision@K / Recall@K
University recommendation	Precision@K, NDCG, expert relevance rating
Priority recommendation	Agreement with expert / officer decision
API / search performance	Mean and P95 latency
Usability	Task completion rate, time, user satisfaction

20. Comparative Experiment

A small experiment should compare simple and advanced approaches so the project includes a clear research contribution.

Method	Description	Example Precision@5
A. Keyword Matching	Traditional keyword overlap	62%
B. Embedding Similarity	Semantic vector similarity	78%
C. Hybrid Matching	Embedding + capability scoring	89%

Research question: Does hybrid semantic matching combined with structured university capability scoring produce more relevant recommendations than keyword-only or embedding-only matching?

21. Why This Version Has Higher Capstone Value

Basic Portal View	Enhanced Capstone View
Complaint submission	Structured AI problem understanding
Simple similarity	Semantic clustering of repeated societal problems
AI says High/Medium/Low	Explainable weighted priority intelligence
University recommendation	Explainable hybrid recommendation algorithm
Project completed	Innovation Readiness Level + measurable deployment
Dashboard counts	Geospatial intelligence + impact analytics
AI feature demo	Measured AI evaluation and comparative experiment
User actions	Auditable human-in-the-loop decisions

22. Innovation & Future Scope

The main innovation is not simply creating another complaint portal. It is creating an explainable, measurable problem-to-innovation pipeline.

- A citizen problem can become a structured, validated innovation challenge.
- Repeated reports can be recognized as one larger societal problem cluster.
- A challenge can be intelligently matched with the most capable university.
- A university project can receive industry mentorship, technology or funding.
- A successful prototype can become a field pilot and deployable solution.
- The entire journey, decisions and social impact remain visible to stakeholders.

Future extensions

- Hindi and regional-language problem submission.
- Mobile/PWA and offline-first submission for low-connectivity areas.
- Advanced recommendation learning from previous successful projects.
- Integration with existing government grievance/public-service systems where permitted.
- Startup/incubation pipeline for promising solutions.
- Research papers, patents and technology-transfer tracking.
- Expansion from Jharkhand to other states / national deployment.

23. Expected Impact

Stakeholder	Expected Impact
Citizens	Transparent path from reporting a problem to seeing validated action and measurable results.
Students	Real-world projects, practical learning and innovation opportunities.
Universities	Demand-driven research and stronger community / industry engagement.
Industry	Access to real problems, talent, pilots and collaboration opportunities.
Government	Centralized visibility, explainable prioritization, auditability and measurable outcomes.
Society	More local problems can receive technically appropriate and trackable solutions.

24. One-Line Pitch

We are building an explainable AI-powered platform that turns real problems reported by communities into collaborative innovation projects by understanding and clustering the problems, matching them with the right universities and industry partners, and tracking each solution all the way to measurable real-world impact.

25. Capstone Quality Target

The project should be evaluated as a strong final-year capstone only if the core intelligence modules are implemented and measured - not merely presented as UI mockups. The target is a polished end-to-end system with explainable algorithms, measurable AI performance, secure role-based workflows, strong demo quality and a clear societal impact story.

Area	Target Quality
Problem relevance	Excellent
Technical complexity	Excellent
AI depth	Excellent
Innovation	Excellent
Research contribution	Strong
Security & auditability	Strong
Demo potential	Excellent
Evaluation methodology	Strong

26. Final Demonstration Checklist

- Submit a real problem with media and location.
- Validate it as a government user and show the audit event.
- Show structured AI extraction, classification confidence and evidence.
- Show a semantic duplicate cluster with similarity scores.
- Show explainable priority calculation and a human override option.
- Show top university recommendations with weighted score breakdowns.
- Create a university team, proposal and milestones with IRL progression.
- Connect an industry collaborator and record the contribution.
- Record pilot/deployment results and calculate impact indicators.
- Finish with the government heatmap/analytics dashboard and AI evaluation metrics.

Capstone goal: demonstrate one complete, intelligent and measurable problem-to-impact journey rather than many disconnected features.